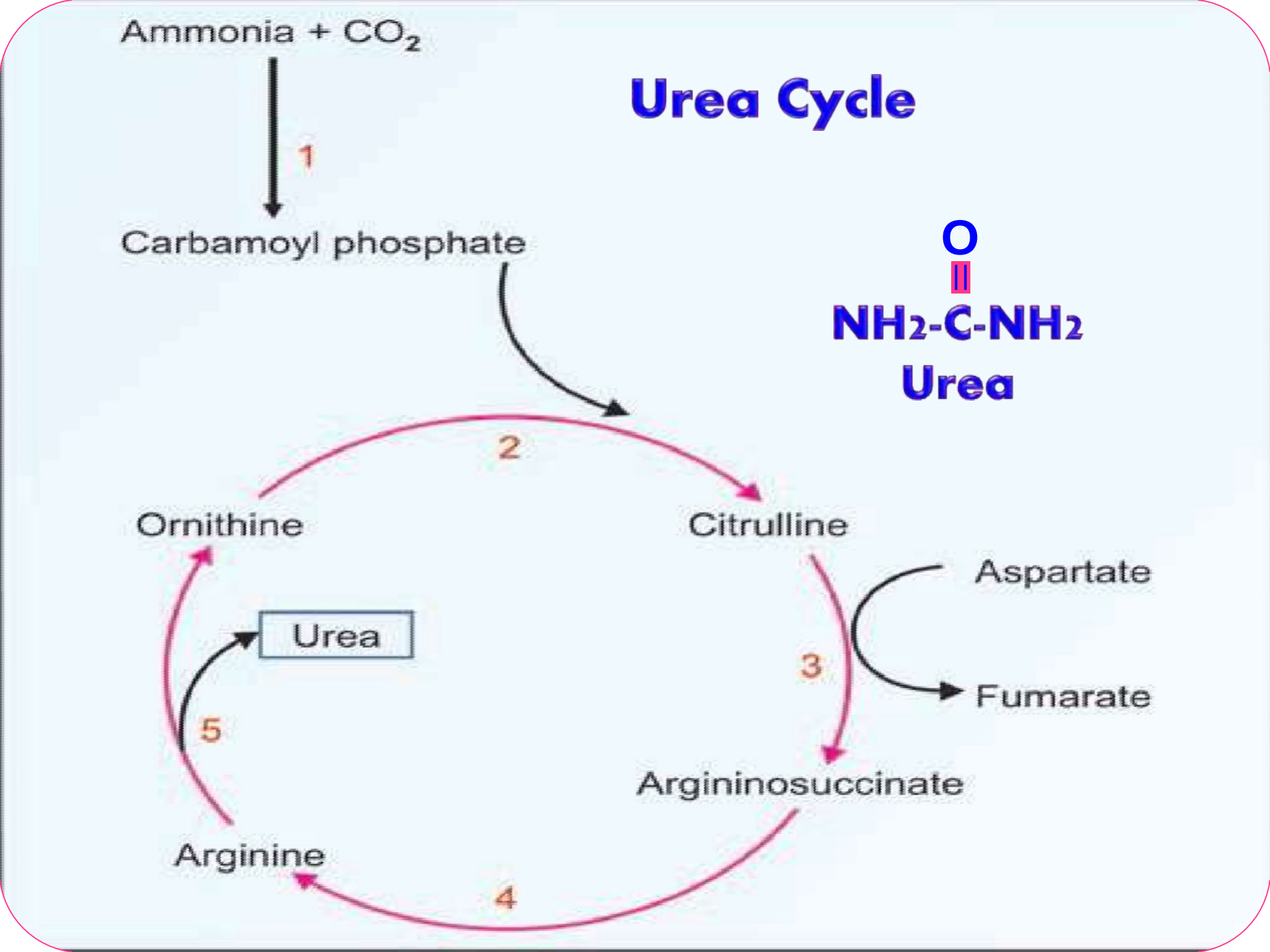
Urea Cycle

Urea Cycle

- The urea cycle is the first metabolic pathway to be elucidated.
- The cycle is known as Krebs–Henseleit urea cycle.
- Ornithine is the first member of the reaction, it is also called as Ornithine cycle.
- Urea is synthesized in liver & transported to kidneys for excretion in urine.

- The two nitrogen atoms of urea are derived from two different sources, one from ammonia & the other directly from the a-amino group of aspartic acid.
- Carbon atom is supplied by CO_2
- Urea is the end product of protein metabolism (amino acid metabolism).

- Urea accounts for 80-90% of the nitrogen containing substances excreted in urine.
- Urea synthesis is a five-step cyclic process, with five distinct enzymes.
- The first two enzymes are present in mitochondria while the rest are localized in cytosol.

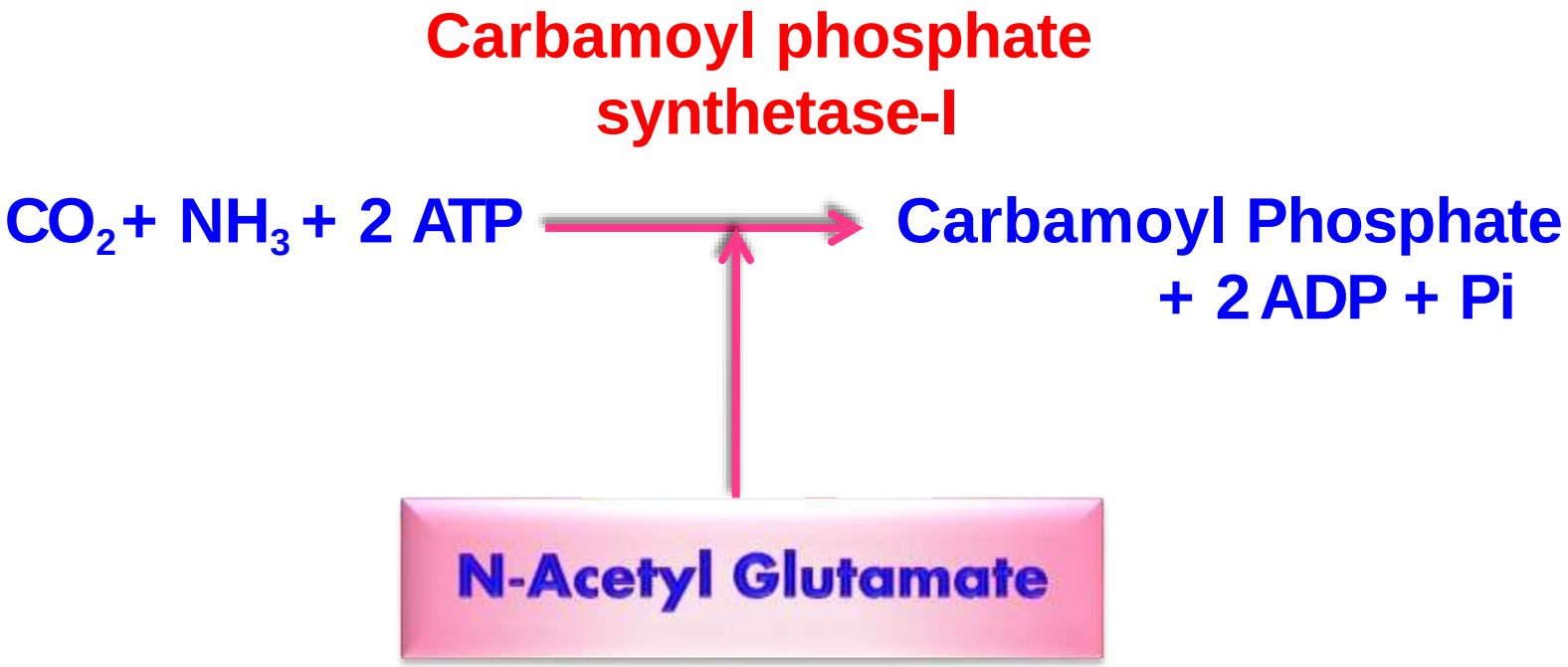

Ammonia + CO_2
1
Carbamoyl phosphate
Urea Cycle
O
II
NH_2-C-NH_2
Urea
2
Ornithine
Citrulline
Aspartate
3
Fumarate
Urea
5
Argininosuccinate
Arginine
4

Step: 1 Formation of carbamoyl phosphate

- **Carbamoyl phosphate synthase I (CPS I) of mitochondria catalyses the condensation of NH_4^+ ions with CO_2 to form carbamoyl phosphate.**
- **This step consumes two ATP & is irreversible.**
- **It is a rate-limiting.**

- **CPS I requires N-acetylglutamate for its activity.**
- **Carbamoyl phosphate synthase II (CPS II) - involved in pyrimidine synthesis & it is present in cytosol.**
- **It accepts amino group from glutamine & does not require N-acetylglutamate for its activity.**

Step: 1 Formation of carbamoyl phosphate



Carbamoyl Phosphate Synthetases

CPS-I

- Mitochondria
- Uses NH_3
- Urea Cycle
- Activated – NAG

CPS-II

- Cytosol
- Uses Glutamine
- Pyrimidine biosynthesis
- Inhibited - CTP

Step 2: Formation of Citrulline

- The second reaction is also mitochondrial.
- Citrulline is synthesized from carbamoyl phosphate & ornithine by ornithine transcarbamoylase.
- Ornithine is regenerated & used in urea cycle.

- **Ornithine & citrulline are basic amino acids. (Never found in protein structure due to lack of codons).**
- **Citrulline is transported to cytosol by a transporter system.**
- **Citrulline is neither present in tissue proteins nor in blood; but it is present in milk.**

Step 2: Formation of Citrulline

Ornithine + Carbamoyl phosphate —(Ornithine Transcarbomylase)→ Citrulline + Pi

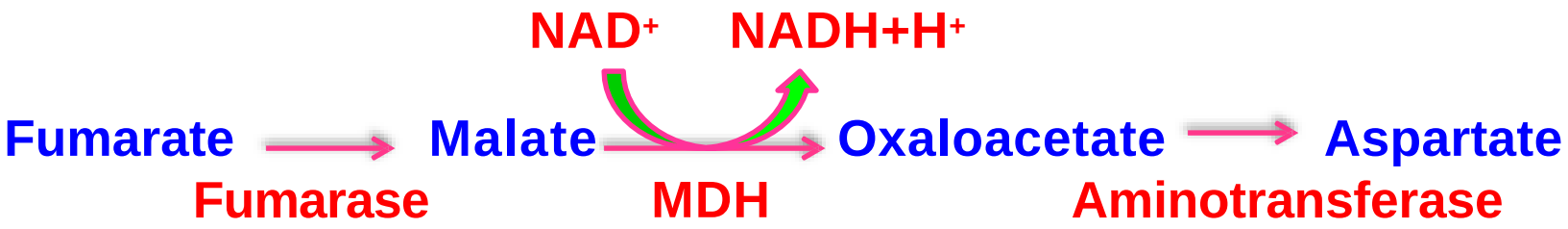
Step 3: Formation of Arginosuccinate

- Citrulline condenses with aspartate to form arginosuccinate by the enzyme Arginosuccinate synthetase.
- Second amino group of urea is incorporated.
- It requires ATP, it is cleaved to AMP & PPi
- 2 High energy bonds are required.
- Immediately broken down to inorganic phosphate (Pi).

Step: 4 Formation of Arginine or cleavage of Arginosuccinate

- The enzyme Argininosuccinase or argininosuccinate lyase cleaves arginosuccinate to arginine & fumarate (an intermediate in TCA cycle)
- Fumarate provides connecting link with TCA cycle or gluconeogenesis.

- The fumarate is converted to oxaloacetate via fumarase & MDH & transaminated to aspartate.
- **Aspartate is regenerated in this reaction.**

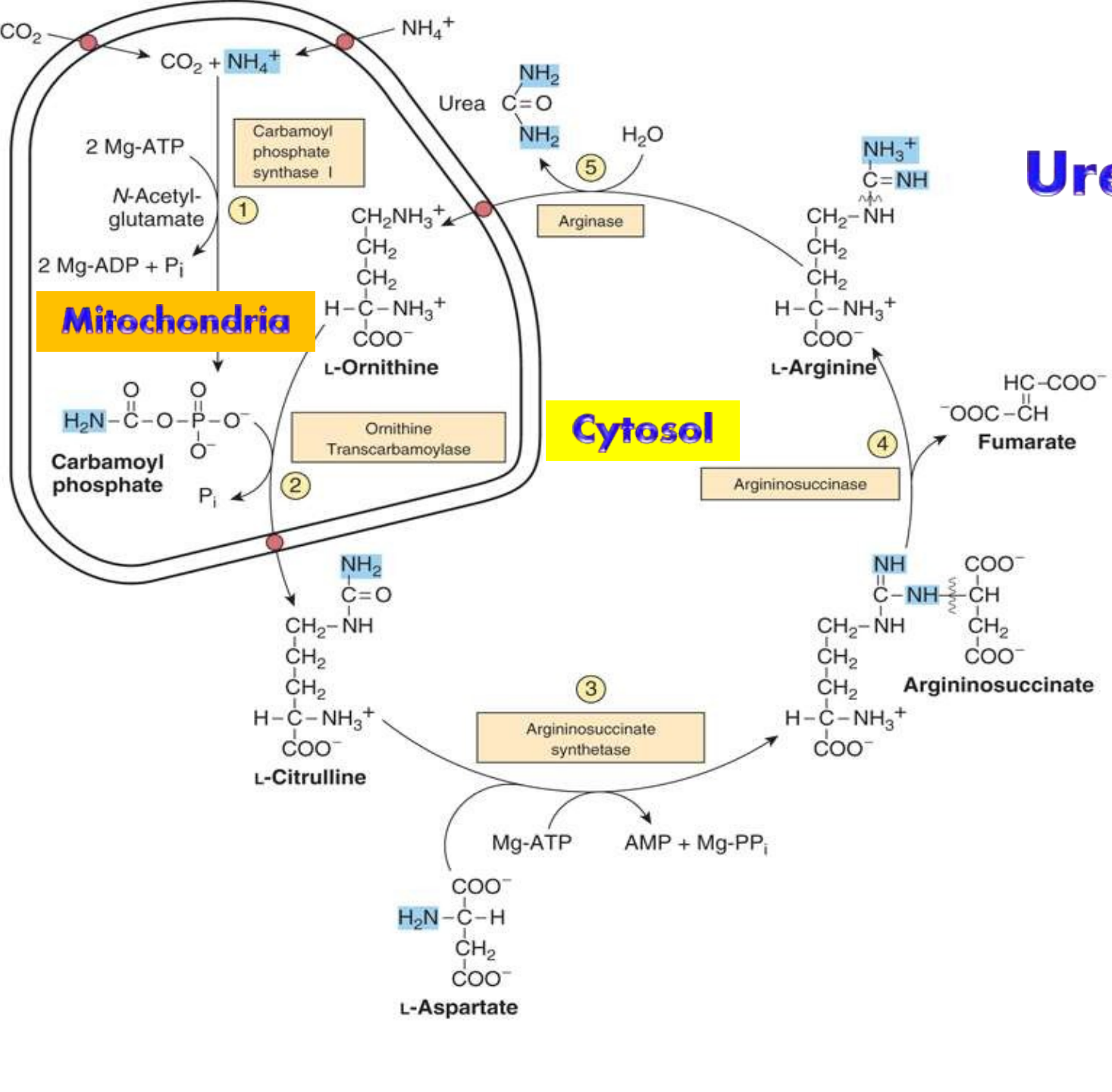


Step 5: Formation of Urea

- **Arginase is the 5th and final enzyme that cleaves arginine to yield urea & ornithine.**
- **Ornithine is regenerated, enters mitochondria for its reuse in the urea cycle.**
- **Arginase is activated by Co^{2+} & Mn^{2+}**
- **Ornithine & lysine compete with arginine (competitive inhibition).**

- Arginase is mostly found in the liver, while the rest of the enzymes (four) of urea cycle are also present in other tissues.
- Arginine synthesis may occur to varying degrees in many tissues.
- But only the liver can ultimately produce urea.

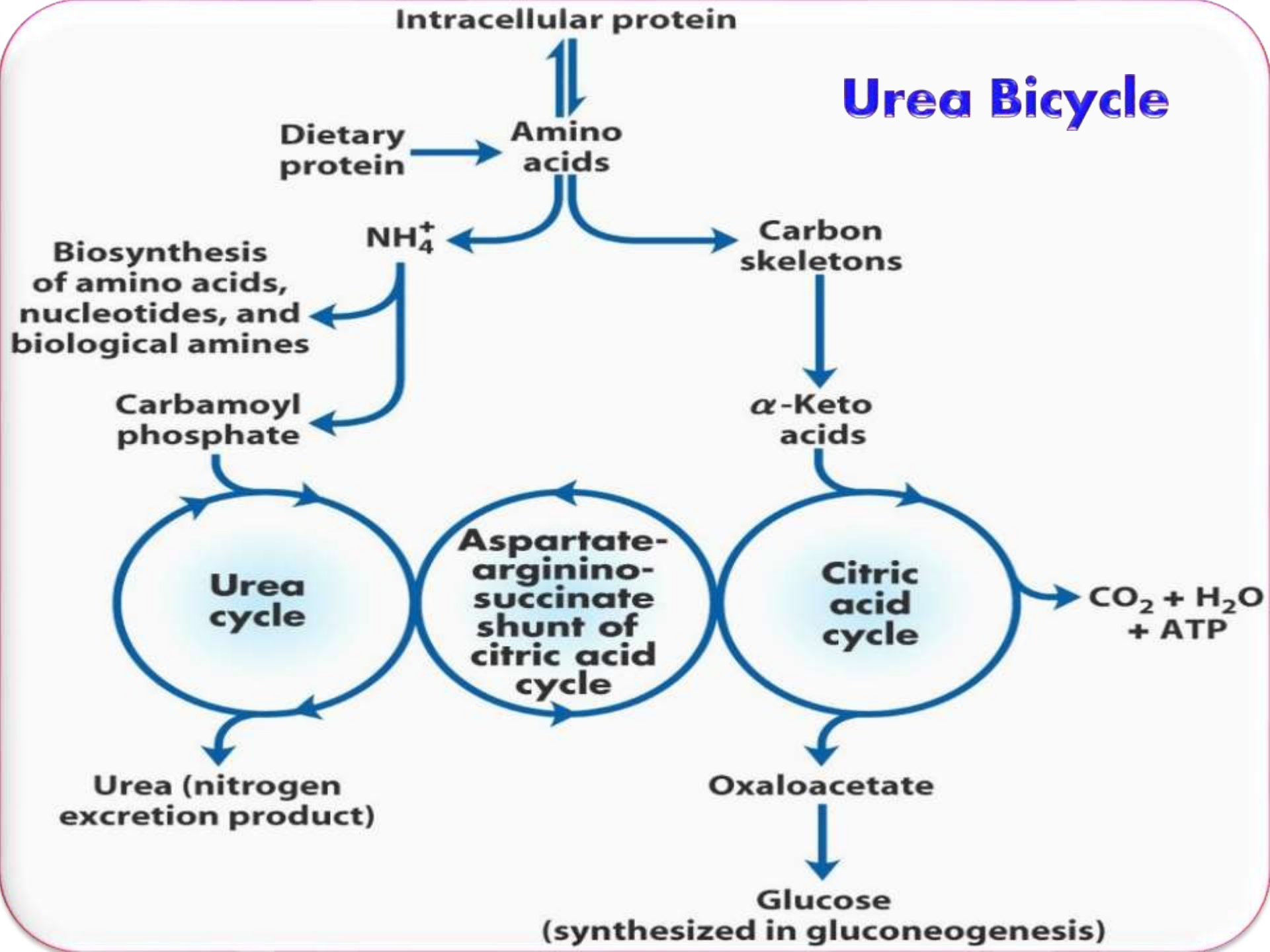
Urea Cycle


CO_2
NH_4^+
$CO_2 + NH_4^+$
2 Mg-ATP
Carbamoyl phosphate synthase I
N-Acetyl-glutamate
1
2 Mg-ADP + P_i
Mitochondria
$CH_2NH_3^+$
L-Ornithine
Carbamoyl phosphate
Ornithine Transcarbamoylase
P_i
2
Cytosol
Urea
NH_2
C=O
NH_2
H_2O
5
Arginase
NH_3^+
C=NH
L-Arginine
HC–COO⁻
⁻OOC–CH
Fumarate
4
Argininosuccinase
Argininosuccinate
3
Argininosuccinate synthetase
L-Citrulline
Mg-ATP
AMP + Mg-PP_i
L-Aspartate

Energetics of Urea Cycle

- The overall reaction may be summarized as:
- $NH_3 + CO_2 + Aspartate \rightarrow Urea + fumarate$
- 2ATPs are used in the 1st reaction.
- Another ATP is converted to AMP + PPi in the 3rd step, which is equivalent to 2 ATPs.
- The urea cycle consumes 4 high energy phosphate bonds.
- Fumarate formed in the 4th step may be converted to malate.

- Malate when oxidised to oxaloacetate produces 1 NADH equivalent to 2.5 ATP.
- So net energy expenditure is only 1.5 high energy phosphates.
- The urea cycle & TCA cycle are interlinked & it is called as "urea bicycle".


Urea Bicycle
Intracellular protein
Dietary protein
Amino acids
NH_4^+
Biosynthesis of amino acids, nucleotides, and biological amines
Carbon skeletons
Carbamoyl phosphate
α-Keto acids
Urea cycle
Aspartate-arginino-succinate shunt of citric acid cycle
Citric acid cycle
$CO_2 + H_2O$ + ATP
Urea (nitrogen excretion product)
Oxaloacetate
Glucose (synthesized in gluconeogenesis)

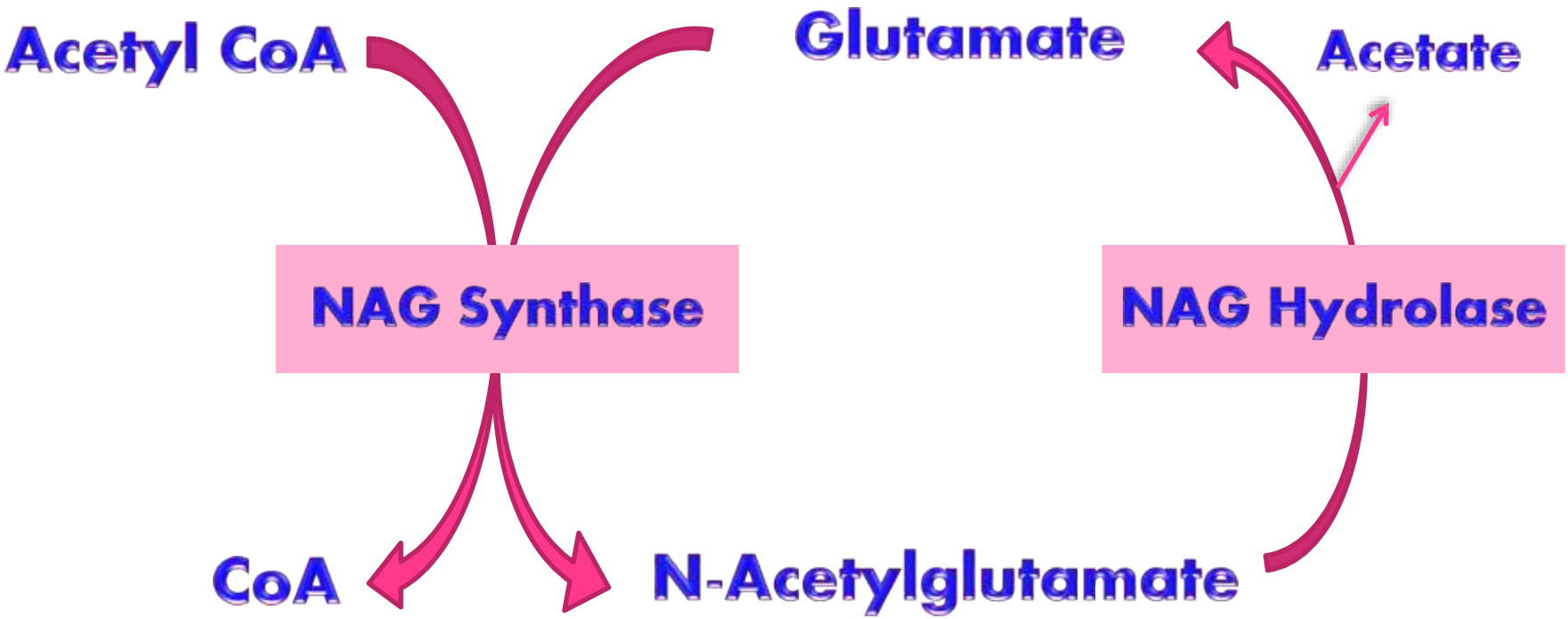
Significance of Urea Cycle

- Toxic ammonia is converted into less-toxic urea.
- Synthesis of semi-essential amino acid-arginine.
- Ornithine is precursor of Proline, Polyamines.
- Polyamines include putrescine, spermidine, spermine.
- Polyamines have diverse roles in cell growth & proliferation.

Regulation of Urea Cycle

- Carbamoyl phosphate synthase (CPS-I) is rate limiting enzyme in urea cycle.
- CPS-I is allosterically activated by N-acetylglutamate (NAG).
- It is synthesized from glutamate & acetyl CoA by synthase & degraded by a hydrolase.
- The rate of urea synthesis in liver is correlated with the concentration of N-acetylglutamate.

Formation & degradation of N-acetylglutamate



- High concentrations of arginine increase NAG.
- The consumption of a protein-rich meal increases the level of NAG in liver, leading to enhanced urea synthesis.
- CPS-I & GDH are present in mitochondria.
- They coordinate with each other in the formation of NH3 & its utilization for carbamoyl phosphate synthesis.

Compartmentalization

- 1ST two enzymes – Mitochondria.
- Fumarate inhibits 4th step.
- Fumarase - in mitochondria.
- Argininosuccinate lyase – in cytoplasm.

Disposal of urea

- Urea produced in the liver freely diffuses & is transported in blood to kidneys & excreted.
- A small amount of urea enters the intestine where it is broken down to CO_2 & NH_3 by the bacterial enzyme urease.
- This ammonia is either lost in the feces or absorbed into the blood.

Disorders of the Urea Cycle

- **The main function of Urea cycle is to remove toxic ammonia from blood as urea.**
- **Defects in the metabolism of conversion of ammonia to urea, i.e., Urea cycle leads to Hyperammonaemia or NH_3 intoxication.**

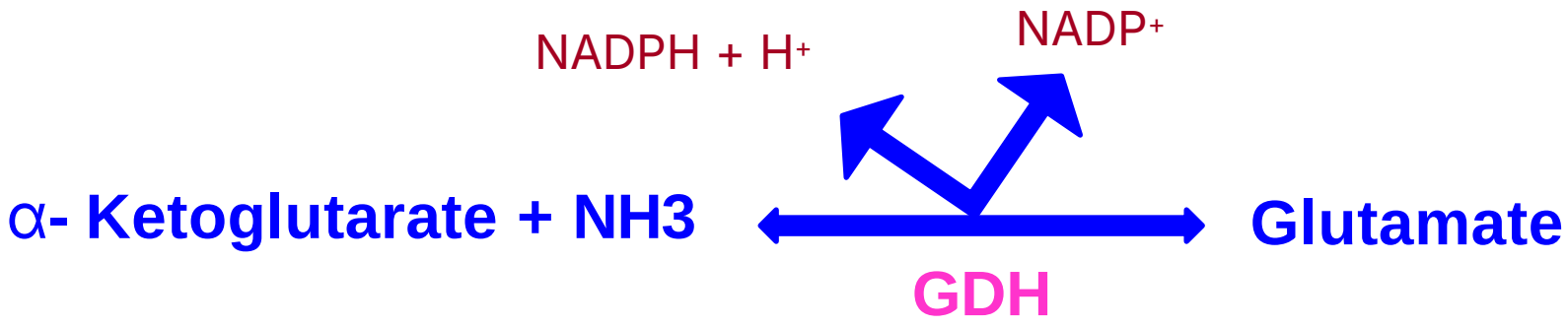
Hyperammonaemia

- **Inherited disorders of urea cycle enzymes- familial hyperammonaemia.**
- **Acquired disorders- Liver Disease, severe Renal disease - Acquired hyperammonaemia.**

Ammonia toxicity

- Increased levels of ammonia crosses BBB, formation of glutamate.

- More utilization of α-ketoglutarate.
- Decreased levels of α- Ketoglutarate in Brain.
- α-KG is a key intermediate in TCA cycle.
- Decreased levels impairs TCA cycle.
- Decreased ATP production.



Hepatic Coma (Acquired Hyperammonemia)

- In diseases of the liver, hepatic failure can finally lead to hepatic coma & death.
- Hyperammonemia is the characteristic feature of liver failure.
- The condition is also known as portal systemic encephalopathy.

- **Normally the ammonia & other toxic compounds produced by intestinal bacterial metabolism are transported to liver by portal circulation & detoxified by the liver.**
- **But when there is portal systemic shunting of blood, the toxins bypass the liver & their concentration in systemic circulation rises.**

Signs, symptoms & treatment

- CNS dysfunction or manifestations of failure of liver function (ascites, jaundice, hepatomegaly, edema, hemorrhage).
- The management of the condition is difficult.
- A low protein diet & intestinal disinfection (bowel clearing & antibiotics), withholding hepatotoxic drugs & maintenance of electrolyte & acid-base balance.

Inherited disorders of urea cycle

Disoeders	Defective Enzyme	Products accumulated
Hyperammonaemia-1	Carbamoyl Phosphate Syntethase -1	Ammonia
Hyperammonaemia-2	Ornithine transcarbomylase (orotic aciduria-most common)	Ammonia
Citrullinemia	Argininino succinate Syntheatse	Citrulline
Arginosuccinic aciduria	Argininosuccinat e lyase	Arginosuccinate
Argininemia	Arginase	Arginine

N-Acetyl Glutamate synthase deficiency

- **Autosomal Recessive.**
- **A severe neonatal disorder with fatal consequences.**
- **Treatment with structural analog N-carbamoyl-L-glutamate – activates CPS-I.**
- **Ornithine Transporter Deficiency (ORNT1 gene):**
- **Ornithine is accumulated in Cytoplasm.**
- **HHH syndrome – Hyper-ornithinemia, Hyper-ammonemia, Homocitrillinuria.**

Symptoms

- Increased levels of ammonia results in
- Slurring of speech
- Blurring of the vision
- Convulsions
- Nausea, Vomiting
- Neurological Deficits
- Mental Retardation
- Coma & Death.

Diagnosis

- Increased levels of ammonia in blood & urine.
- Increased glutamine – in CSF, excreted in urine.
- Decreased blood urea levels.
- Urea cycle intermediates accumulate in blood & excreted in urine.

Treatment

- Intravenous administration of sodium benzoate, phenyllacetate.
- These condense with glycine & glutamate to form water soluble products that can be easily excreted.
- By this, ammonia can be trapped & removed from the body.
- In toxic hyperammonemia, hemodialysis may become necessary.

Blood urea significance

- Normal blood urea concentration is 10-40 mg/dl.
- About 15-30 g of urea (7-15 g nitrogen) is excreted in urine per day.
- Blood urea estimation is a screening test for the evaluation of kidney (renal) function.
- Elevation in blood urea may be broadly classified into three categories.

Pre-renal

- This is associated with increased protein breakdown, leading to a negative nitrogen balance.
- Observed after major surgery, prolonged fever, diabetic coma, thyrotoxicosis etc.
- In leukemia & bleeding disorders also, blood urea is elevated.

Renal

- In renal disorders like acute glomerulonephritis, chronic nephritis, nephrosclerosis, polycystic kidney, blood urea is increased.
- Post-renal:
- Due to obstruction in the urinary tract (e.g. tumors, stones, enlargement of prostate gland etc.) blood urea is elevated.
- This is due to increased reabsorption of urea from the renal tubules.

References

- Chatterjea
- **Lippincott & Harper**